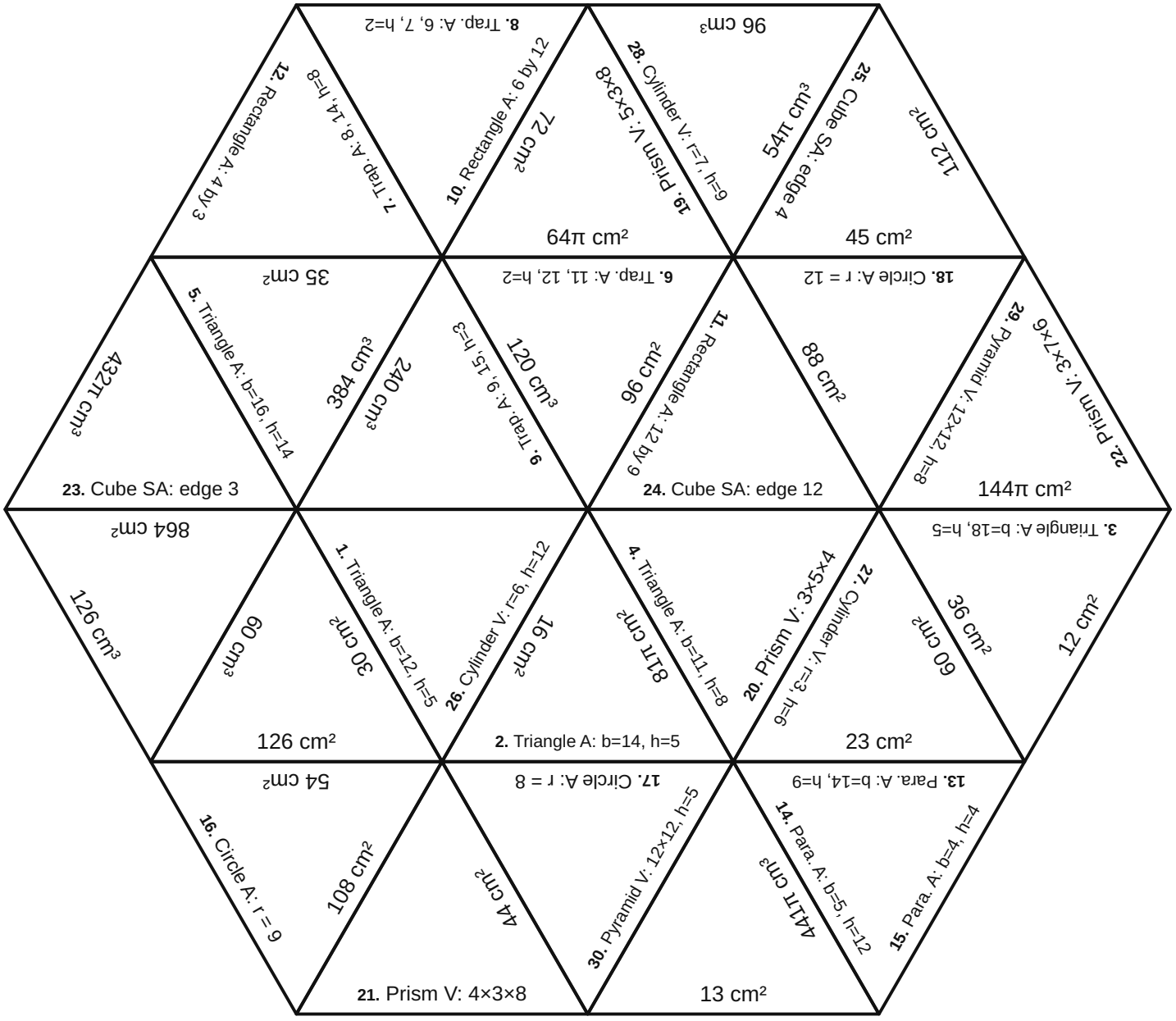
# Area and Volume Tarsia

Mr. K's Classroom · Hexagon of 24 triangles, 30 matches Name: _____

**Directions:** Find each area or volume. Leave answers in terms of π where it appears. Cut out the 24 triangles. Match each problem to its answer so that touching edges fit, and build a **hexagon**. Glue it on a sheet of paper. Show your work for every problem on the work pages (the number before each problem tells you where).



# Show Your Work

Area and Volume Tarsia · Problems 1–15 Name: _____

| | | |
|---|---|---|
| **1.** Triangle A: b=12, h=5 | **2.** Triangle A: b=14, h=5 | **3.** Triangle A: b=18, h=5 |
| **4.** Triangle A: b=11, h=8 | **5.** Triangle A: b=16, h=14 | **6.** Trap. A: 11, 12, h=2 |
| **7.** Trap. A: 8, 14, h=8 | **8.** Trap. A: 6, 7, h=2 | **9.** Trap. A: 9, 15, h=3 |
| **10.** Rectangle A: 6 by 12 | **11.** Rectangle A: 12 by 9 | **12.** Rectangle A: 4 by 3 |
| **13.** Para. A: b=14, h=9 | **14.** Para. A: b=5, h=12 | **15.** Para. A: b=4, h=4 |

# Show Your Work

Area and Volume Tarsia · Problems 16–30 Name: _____

| | | |
|---|---|---|
| **16.**<br>Circle A: r = 9 | **17.**<br>Circle A: r = 8 | **18.**<br>Circle A: r = 12 |
| **19.**<br>Prism V: 5×3×8 | **20.**<br>Prism V: 3×5×4 | **21.**<br>Prism V: 4×3×8 |
| **22.**<br>Prism V: 3×7×6 | **23.**<br>Cube SA: edge 3 | **24.**<br>Cube SA: edge 12 |
| **25.**<br>Cube SA: edge 4 | **26.**<br>Cylinder V: r=6, h=12 | **27.**<br>Cylinder V: r=3, h=6 |
| **28.**<br>Cylinder V: r=7, h=9 | **29.**<br>Pyramid V: 12×12, h=8 | **30.**<br>Pyramid V: 12×12, h=5 |